

DOTHAN, AL, USA

WMO: 722268

Lat: 31.317N

Lon: 85.450W

Elev: 374

StdP: 14.50

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13839

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.2	30.5	11.5	10.0	35.7	16.1	12.6	38.7	24.4	54.5	21.9	50.0	7.6	320	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	96.5	75.5	94.0	75.3	92.1	75.0	79.9	89.3	78.8	87.9	77.9	86.7	7.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	144.7	83.1	76.6	140.5	82.4	75.4	134.8	81.4	43.5	89.6	42.4	88.2	41.5	86.0	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	17.8	15.7	20.9	98.8	4.0	3.0	18.0	100.9	15.7	102.6	13.4	104.3	10.5	106.4
			18.4	81.0	3.8	1.4	15.7	82.0	13.5	82.9	11.3	83.7	8.6	84.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.4	50.2	54.3	60.1	66.8	75.2	80.6	82.2	81.5	77.5	68.4	58.5	52.9
	DBStd	13.38	10.01	9.06	8.14	6.40	5.31	3.89	2.97	3.19	4.64	7.25	8.18	9.40
	HDD50	292	124	54	17	0	0	0	0	0	0	1	18	78
	HDD65	1699	466	314	191	55	5	0	0	0	1	51	228	388
	CDD50	6646	130	173	330	505	780	919	997	976	825	571	272	168
	CDD65	2578	7	13	38	110	320	469	532	511	377	156	32	13
	CDH74	23650	18	62	284	916	2956	4594	5297	4910	3184	1215	175	39
	CDH80	9938	0	3	32	217	1202	2154	2429	2206	1333	344	17	1
(14) Wind	WSAvg	7.2	8.3	8.4	8.5	8.0	7.2	6.5	5.9	5.6	6.3	6.8	7.0	7.7
(15) Precipitation	PrecAvg	53.3	5.0	4.4	5.5	3.9	2.8	5.2	6.0	4.4	4.0	2.9	4.2	3.8
	PrecMax	64.7	12.8	8.5	11.5	9.6	6.9	10.5	18.7	9.9	14.7	6.5	11.1	8.5
	PrecMin	40.5	1.0	1.2	0.2	0.9	0.0	1.8	2.5	1.3	1.2	0.0	1.5	1.4
	PrecStd	8.0	2.8	2.0	3.1	2.3	1.8	2.8	3.6	2.4	3.4	1.9	2.4	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.4	79.4	83.5	88.2	95.3	99.7	99.0	98.6	95.9	89.9	82.1	78.2
		MCWB	65.8	65.9	65.4	70.1	71.9	76.3	76.3	77.1	73.6	72.8	68.5	67.6
	2%	DB	72.6	75.5	80.6	84.9	91.5	96.5	95.2	94.8	92.6	86.6	78.9	74.1
		MCWB	64.1	64.2	64.6	68.1	71.2	75.1	76.5	77.2	73.5	70.9	66.8	66.3
	5%	DB	69.7	72.3	77.3	82.3	89.4	93.3	93.0	92.6	90.2	83.9	75.4	71.0
		MCWB	63.3	62.8	63.5	66.7	70.5	74.1	76.3	76.7	73.9	69.5	64.9	65.4
	10%	DB	65.8	68.7	74.1	79.9	86.7	90.6	90.9	90.4	87.8	81.1	72.6	67.8
		MCWB	60.7	61.1	61.6	65.3	69.7	73.8	76.1	76.4	73.5	68.0	64.2	63.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.8	69.2	70.5	74.0	77.1	80.3	81.4	81.5	78.6	77.4	73.2	71.2
		MCDB	71.6	73.8	75.6	81.7	86.4	91.4	89.9	90.9	86.3	83.3	76.9	74.4
	2%	WB	66.8	67.4	68.5	71.9	75.0	78.7	79.5	80.0	77.4	75.1	70.8	68.8
		MCDB	70.7	73.0	74.1	79.6	83.5	89.6	89.4	88.8	85.1	80.9	74.3	71.6
	5%	WB	64.4	65.5	66.8	70.2	73.7	77.5	78.4	78.9	76.4	73.2	68.5	66.6
		MCDB	67.9	70.0	73.1	77.3	82.0	87.1	87.8	87.4	84.4	79.2	73.2	69.8
	10%	WB	61.9	63.2	64.9	68.6	72.6	76.3	77.6	78.0	75.5	71.6	66.0	64.0
		MCDB	65.5	67.0	71.3	75.5	81.0	85.2	86.6	86.3	83.7	77.6	71.1	67.3
(35) Mean Daily Temperature Range	5% DB	MDBR	20.0	20.2	21.7	21.8	21.0	18.4	17.8	17.6	18.9	21.5	21.9	19.0
		MCDBR	20.4	21.9	23.9	23.7	22.9	22.1	20.2	20.1	21.9	22.6	22.3	19.6
		MCWBR	13.5	12.6	11.2	9.5	6.9	6.0	5.9	6.4	7.3	9.3	12.0	12.9
	5% WB	MCDBR	18.2	18.2	19.1	18.7	18.7	17.9	17.8	17.6	17.5	17.1	17.5	17.5
		MCWBR	13.9	12.5	11.5	9.9	7.2	6.6	6.4	6.4	6.9	8.8	11.6	13.7
(40) Clear-Sky Solar Irradiance	taub		0.323	0.332	0.354	0.381	0.431	0.470	0.508	0.498	0.428	0.370	0.346	0.329
	taud		2.520	2.494	2.436	2.367	2.256	2.212	2.116	2.140	2.316	2.452	2.466	2.521
	Ebn at Noon		287	295	294	289	273	262	251	252	267	278	276	279
	Edh at Noon		27	31	35	39	44	46	50	48	39	32	29	26
(44) All-Sky Solar Radiation	RadAvg		951	1135	1517	1858	2033	1930	1886	1753	1582	1373	1069	826
	RadStd		58	141	124	152	144	148	118	116	121	150	89	88

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		+0.88	N/A	N/A	N/A	N/A	N/A	N/A	-167	+299	+196

Nomenclature: See separate page